

# Practice Problems 2: Lectures 7–11

BUS 411 — Fixed Income Security Analysis and Valuation

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## Instructions

These 20 numerical problems review Lectures 7–11. Four problems are provided for each lecture. Several are variants of Assignment 2 questions, while others extend worked examples from the lecture notes. Attempt each problem before opening its solution. Keep rates in decimal form during calculations and state monetary units.

## 1 Lecture 7: Interest-Rate Models

### 1.1 Problem 1: Historical Short-Rate Volatility

Weekly short-rate changes, in basis points, are

$$2, -2, 4, -1, 3, -3, 5, 0.$$

- Compute the sample mean.
- Compute the sample weekly standard deviation.
- Annualize the standard deviation using  $\sqrt{52}$ .
- Interpret the annualized result and identify one limitation.

#### Solution

There are eight observations. Their sum is 8 bps, so

$$\bar{x} = \frac{8}{8} = \boxed{1.00 \text{ bp per week}}.$$

The deviations from the mean are 1, −3, 3, −2, 2, −4, 4, −1. Their squared deviations sum to

$$1 + 9 + 9 + 4 + 4 + 16 + 16 + 1 = 60.$$

Using the sample denominator  $n - 1 = 7$ ,

$$s = \sqrt{\frac{60}{7}} = \boxed{2.9277 \text{ bps per week}}.$$

The square-root-of-time annualization is

$$s_{\text{annual}} = 2.9277\sqrt{52} = \boxed{21.11 \text{ bps per year}}.$$

This describes the annualized scale of short-rate changes if weekly changes are independent and have stable variance. Those assumptions may fail when volatility clusters or the monetary-policy regime changes.

## 1.2 Problem 2: Mean Reversion and a One-Standard-Deviation Shock

The short rate follows the Euler approximation

$$\Delta r = a(\theta - r)\Delta t + \sigma\sqrt{\Delta t}\varepsilon,$$

where  $r = 5.00\%$ ,  $\theta = 3.50\%$ ,  $a = 0.80$ ,  $\sigma = 1.20\%$  per year,  $\Delta t = 0.25$ , and  $E[\varepsilon] = 0$ .

- Compute the expected quarterly rate change.
- Compute the expected rate after one quarter.
- Compute the standard deviation of the quarterly shock.
- Give the rates corresponding to shocks of  $\varepsilon = +1$  and  $-1$ .

### Solution

The expected shock is zero. The mean-reversion component is

$$E[\Delta r] = 0.80(0.035 - 0.050)(0.25) = -0.0030 = -30 \text{ bps.}$$

Therefore,

$$E[r_{t+0.25}] = 0.0500 - 0.0030 = \boxed{4.70\%}.$$

The shock standard deviation is

$$\sigma\sqrt{\Delta t} = 0.012\sqrt{0.25} = 0.0060 = 60 \text{ bps.}$$

For  $\varepsilon = +1$ , the rate is  $4.70\% + 0.60\% = \boxed{5.30\%}$ ; for  $\varepsilon = -1$ , it is  $4.70\% - 0.60\% = \boxed{4.10\%}$ . Mean reversion controls the center of the distribution, while  $\sigma$  controls its dispersion.

## 1.3 Problem 3: Nodes in a Lognormal Short-Rate Tree

A model specifies year-2 short rates as

$$r_{2,j} = a_2 e^{(2j-2)\sigma}, \quad j = 0, 1, 2,$$

with  $a_2 = 4.00\%$  and  $\sigma = 0.15$ .

- Compute all three year-2 short rates.
- Compute the one-year discount factor  $1/(1+r)$  at each node.
- Which parameter sets the middle level of the rates?

- d. If  $\sigma$  rises while  $a_2$  is unchanged, what happens numerically to the distance between the low and high nodes?

#### **i** Solution

For  $j = 0, 1, 2$ , the exponents are  $-0.30, 0, +0.30$ . Thus,

$$r_{2,0} = 0.04e^{-0.30} = \boxed{2.9633\%},$$

$$r_{2,1} = 0.04 = \boxed{4.0000\%},$$

$$r_{2,2} = 0.04e^{0.30} = \boxed{5.3994\%}.$$

The associated one-year discount factors are approximately

$$\boxed{0.9712, \quad 0.9615, \quad 0.9488}.$$

The level parameter  $a_2$  fixes the middle node. The current low-to-high distance is  $5.3994\% - 2.9633\% = 2.4361\%$ . A larger  $\sigma$  lowers the low node and raises the high node, widening this distance; this is how volatility controls dispersion.

### 1.4 Problem 4: A One-Period Bond-Option Tree

One year from today, a zero-coupon bond will be worth either 94 in the up-rate state or 98 in the down-rate state. A European call expiring then has strike 96. The risk-neutral probability of the up-rate state is 0.55, and today's one-year short rate is 4%.

- Compute the call payoff in each state.
- Compute the call's time-0 value.
- Compute the hedge ratio  $(C_d - C_u)/(B_d - B_u)$ .
- Explain why the call pays more in the down-rate state.

#### **i** Solution

The payoffs are

$$C_u = \max(94 - 96, 0) = 0, \quad C_d = \max(98 - 96, 0) = 2.$$

Risk-neutral valuation gives

$$C_0 = \frac{0.55(0) + 0.45(2)}{1.04} = \boxed{0.8654}.$$

The option changes by 2 when the bond changes by 4, so

$$h = \frac{2 - 0}{98 - 94} = \boxed{0.50}.$$

When rates fall, discount factors rise and the zero-coupon bond price is higher. The call is therefore in the money in the down-rate state.

## 2 Lecture 8: Bonds with Embedded Options and OAS

### 2.1 Problem 5: Estimating a Static Spread

A corporate bond trades at 101.00. Discounting its promised cash flows at Treasury zero rates plus a trial spread gives:

| Spread | Model value |
|--------|-------------|
| 40 bps | 101.40      |
| 60 bps | 100.72      |

- Explain why the value falls as the spread rises.
- Estimate the static spread by linear interpolation.
- How far is the estimate from each trial spread?
- State what risk the static spread does not model for a callable bond.

#### **i** Solution

A larger spread raises every discount rate, reducing present value. Since 101.00 lies between the two trial values, the spread lies between 40 and 60 bps:

$$s \approx 40 + \frac{101.40 - 101.00}{101.40 - 100.72}(60 - 40) = \boxed{51.76 \text{ bps}}.$$

It is 11.76 bps above 40 and 8.24 bps below 60. Interpolation is an approximation because price is convex rather than perfectly linear in spread. A static spread also ignores the path-dependent exercise of an embedded call; OAS uses a rate lattice to model that behavior.

### 2.2 Problem 6: Pricing a Callable Bond

A two-year annual-pay bond has face value 100 and coupon 6. The time-0 short rate is 5%. At year 1, rates are either 4% or 8%, each with risk-neutral probability 0.5. The bond may be called at 101 at year 1, after the continuation value is calculated. Coupons are added after discounting the next payment.

- Compute the year-2 payment.
- Compute the year-1 option-free hold value at each node.

- c. Apply the call rule and compute the time-0 callable value.
- d. Compute the option-free value and the embedded call value.

#### **i** Solution

The maturity payment is  $100 + 6 = 106$ . The year-1 hold values are

$$V_{1,L} = \frac{106}{1.04} + 6 = 107.9231,$$

$$V_{1,H} = \frac{106}{1.08} + 6 = 104.1481.$$

Both exceed the call price, so the issuer calls in both states and the ex-coupon node value is capped at 101. The callable value is

$$V_0^{\text{call}} = \frac{0.5(101) + 0.5(101)}{1.05} + 6 = \boxed{102.1905}.$$

Without the call,

$$V_0^{\text{straight}} = \frac{0.5(107.9231) + 0.5(104.1481)}{1.05} + 6 = \boxed{106.9863}.$$

Thus, the issuer's embedded call is worth

$$106.9863 - 102.1905 = \boxed{4.7958}$$

to the issuer and reduces the value received by the investor by the same amount.

## 2.3 Problem 7: Pricing a Puttable Bond

A two-year annual-pay bond has face value 100 and coupon 3. The time-0 short rate is 5%. At year 1, rates are either 3% or 10%, each with probability 0.5. The investor may put the bond at 100 at year 1.

- a. Compute the year-2 payment.
- b. Compute the year-1 hold values.
- c. Apply the put rule.
- d. Compute the time-0 puttable and option-free values and the embedded put value.

#### **i** Solution

The maturity payment is  $100 + 3 = 103$ . The hold values are

$$V_{1,L} = \frac{103}{1.03} + 3 = 103.0000,$$

$$V_{1,H} = \frac{103}{1.10} + 3 = 96.6364.$$

The investor keeps the bond in the low-rate state and exercises the put in the high-rate state. Hence the year-1 values are 103 and 100. The time-0 putable value is

$$V_0^{\text{put}} = \frac{0.5(103) + 0.5(100)}{1.05} + 3 = \boxed{99.6667}.$$

The option-free value is

$$V_0^{\text{straight}} = \frac{0.5(103) + 0.5(96.6364)}{1.05} + 3 = \boxed{98.0649}.$$

The embedded investor put is worth

$$99.6667 - 98.0649 = \boxed{1.6017}.$$

## 2.4 Problem 8: OAS, Effective Duration, and Convexity

A callable bond has market price  $P_0 = 100.00$ . Trial OAS prices are 100.40 at 45 bps and 99.95 at 60 bps. After solving for OAS, the model gives  $P_- = 102.60$  when the curve falls 50 bps and  $P_+ = 97.70$  when it rises 50 bps.

- Estimate the OAS.
- Compute effective duration.
- Compute effective convexity.
- Interpret the two measures.

### Solution

Interpolating to the market price,

$$\text{OAS} \approx 45 + \frac{100.40 - 100.00}{100.40 - 99.95}(60 - 45) = \boxed{58.33 \text{ bps}}.$$

With  $\Delta y = 0.005$ ,

$$D_{\text{eff}} = \frac{P_- - P_+}{2P_0\Delta y} = \frac{102.60 - 97.70}{2(100)(0.005)} = \boxed{4.90}.$$

$$C_{\text{eff}} = \frac{P_- + P_+ - 2P_0}{P_0(\Delta y)^2} = \frac{102.60 + 97.70 - 200}{100(0.005)^2} = \boxed{120.00}.$$

Effective duration measures the local price slope after allowing the call decision to change. Effective convexity measures curvature under the same full repricing; both are therefore preferable to cash-flow duration for an option-embedded bond.

### 3 Lecture 9: Interest-Rate Futures

#### 3.1 Problem 9: Treasury Futures Quotation and Margin Cash Flows

A trader buys one Treasury futures contract at 101-12. Subsequent settlements are 101-20, 101-08, and 101-24. Quotes are in 32nds, and one point is worth \$1,000.

- Convert all four prices to decimals.
- Compute each daily cash flow to the long.
- Compute cumulative profit or loss.
- Explain the sign of the cash flows.

#### Solution

The decimal prices are

101.375, 101.625, 101.250, 101.750.

Daily marking to market gives

| Day | Price change | Long cash flow |
|-----|--------------|----------------|
| 1   | +0.250       | +\$250         |
| 2   | −0.375       | −\$375         |
| 3   | +0.500       | +\$500         |

The cumulative result is  $250 - 375 + 500 = \boxed{\$375}$ , equal to  $(101.750 - 101.375)(\$1,000)$ . A long receives cash when the settlement price rises and pays cash when it falls.

#### 3.2 Problem 10: Three-Month SOFR Futures

A long Three-Month SOFR futures position moves from 95.18 to 95.26. Each one-basis- point price move is worth \$25 per contract.

- Compute both implied SOFR rates.
- Compute the price move in basis points.
- Compute the long's dollar gain or loss.
- Explain the inverse price-rate relation.



### **i** Solution

The quoted rate is 100— futures price. Initially it is  $100 - 95.18 = \boxed{4.82\%}$  and afterward it is  $100 - 95.26 = \boxed{4.74\%}$ . The price rose by 0.08, or 8 basis points, so the long earns

$$8(\$25) = \boxed{\$200}.$$

The contract is quoted as 100 minus the implied rate; therefore, an increase in price mechanically corresponds to a decrease in the implied rate.

## 3.3 Problem 11: Cheapest-to-Deliver Bond

A Treasury futures contract settles at 112.50. Ignore accrued interest and financing.

| Bond | Cash price | Conversion factor |
|------|------------|-------------------|
| A    | 119.10     | 1.0550            |
| B    | 111.25     | 0.9850            |
| C    | 106.40     | 0.9450            |

- Compute each invoice price.
- Compute each net delivery cost.
- Identify the cheapest-to-deliver bond.
- Explain why cash price alone is insufficient.

### **i** Solution

Invoice price equals futures price times conversion factor:

| Bond | Invoice price              | Net delivery cost: cash minus invoice |
|------|----------------------------|---------------------------------------|
| A    | $112.50(1.055) = 118.6875$ | 0.4125                                |
| B    | $112.50(0.985) = 110.8125$ | 0.4375                                |
| C    | $112.50(0.945) = 106.3125$ | 0.0875                                |

Bond C is the CTD because it has the smallest net delivery cost. Although C also has the lowest cash price here, the decision is based on cash price relative to the conversion-factor-adjusted invoice price, not cash price by itself.

## 3.4 Problem 12: Futures DV01 and a Portfolio Hedge

Using Bond C from Problem 11, assume modified duration 7.10, clean price 106.40, deliverable face value \$100,000, and conversion factor 0.9450. A long bond portfolio has DV01 of \$30,500.

- Compute the CTD market value.
- Compute CTD DV01.
- Compute approximate futures DV01.
- Compute the contract hedge and state its direction.

#### **i** Solution

The deliverable market value is

$$MV_{CTD} = 1.0640(\$100,000) = \$106,400.$$

Therefore,

$$DV01_{CTD} = 7.10(\$106,400)(0.0001) = \boxed{\$75.5440}.$$

Adjusting for the conversion factor,

$$DV01_F \approx \frac{75.5440}{0.9450} = \boxed{\$79.9407}.$$

The hedge requires

$$N = \frac{30,500}{79.9407} = 381.53.$$

The manager should sell approximately  $\boxed{382}$  futures contracts because the long bond portfolio loses when rates rise, while short futures gain. Rounding to a whole contract leaves a small residual DV01.

## 4 Lecture 10: Credit Exposure and Credit Risk

### 4.1 Problem 13: Spread Duration and DTS

A corporate bond position has market value \$5,000,000, spread duration 4.20, and spread 220 bps. Its spread widens by 30 bps.

- Estimate the percentage price change.
- Estimate the dollar change.
- Compute DTS using spread in percentage points.
- Interpret DTS.

### **i** Solution

Using  $\Delta P/P \approx -SD \Delta s$  and  $\Delta s = 0.0030$ ,

$$\frac{\Delta P}{P} \approx -4.20(0.0030) = \boxed{-1.26\%}.$$

The dollar change is

$$-0.0126(\$5,000,000) = \boxed{-\$63,000}.$$

With spread expressed as 2.20 percentage points,

$$DTS = 4.20(2.20) = \boxed{9.24}.$$

Equivalently, using decimal spread gives 0.0924. DTS combines sensitivity and the current spread level, providing a rough relative measure of spread risk across bonds with different credit quality.

## 4.2 Problem 14: Empirical Duration Multiplier

A \$20 million corporate bond position has analytical duration 6.50 and spread  $s = 2.50\% = 0.025$ . The correlation between Treasury-yield changes and relative spread changes is  $-0.60$ . Their volatilities are  $\sigma_y = 0.25\%$  and  $\sigma_s = 2.00\%$ .

- Compute  $\beta = \rho\sigma_s/\sigma_y$ .
- Compute  $M = 1 + s\beta$  and empirical duration.
- Estimate the loss for a 50 bp Treasury-yield increase.
- Compare it with the analytical-duration loss.

### **i** Solution

$$\beta = -0.60 \frac{0.0200}{0.0025} = \boxed{-4.80}.$$

The duration multiplier and empirical duration are

$$M = 1 + 0.025(-4.80) = \boxed{0.88},$$

$$D_{emp} = 6.50(0.88) = \boxed{5.72}.$$

For  $\Delta y = 0.0050$ , the empirical estimate is

$$\frac{\Delta P}{P} \approx -5.72(0.005) = -2.86\%,$$

$$\Delta P = -0.0286(\$20,000,000) = \boxed{-\$572,000}.$$

Analytical duration gives  $-6.50(0.005) = -3.25\%$ , or  $-\$650,000$ . It overstates this empirically adjusted loss by  $\boxed{\$78,000}$  because spreads are assumed to tighten when Treasury yields rise.

### 4.3 Problem 15: A One-Period Structural Credit Model

A firm owes \$100 million in one year. Its asset value will be \$140 million with probability 0.65 or \$70 million with probability 0.35. Ignore discounting.

- Identify the default state.
- Compute debt and equity payoffs in each state.
- Compute expected debt and equity payoffs.
- Compute expected debt loss and recovery in the default state.

#### **i** Solution

Default occurs when assets are below promised debt, so it occurs in the \$70 million state.

| Asset value | Debt payoff $\min(V_A, 100)$ | Equity payoff $\max(V_A - 100, 0)$ |
|-------------|------------------------------|------------------------------------|
| \$140m      | \$100m                       | \$40m                              |
| \$70m       | \$70m                        | 0                                  |

Expected payoffs are

$$E[D] = 0.65(100) + 0.35(70) = \boxed{\$89.5\text{m}},$$

$$E[E] = 0.65(40) + 0.35(0) = \boxed{\$26.0\text{m}}.$$

Expected debt loss relative to the promised \$100 million is \$10.5 million. Recovery conditional on default is  $70/100 = \boxed{70\%}$ , so loss given default is 30%. Equity is a call option on firm assets with strike equal to debt face value.

### 4.4 Problem 16: Default Intensity and Credit Spread

A reduced-form model has constant annual intensity  $\lambda = 2.5\%$  and loss given default of 60%.

- Compute one-year survival and default probabilities.
- Compute five-year survival and cumulative default probabilities.
- For 100 surviving issuers, compute expected defaults during the next year.

d. Estimate the credit spread using  $s \approx \lambda \times LGD$ .

#### **i** Solution

For constant intensity,  $S(t) = e^{-\lambda t}$ . At one year,

$$S(1) = e^{-0.025} = 0.97531,$$

$$PD(1) = 1 - S(1) = \boxed{2.4690\%}.$$

At five years,

$$S(5) = e^{-0.125} = \boxed{88.2497\%},$$

$$PD(5) = 1 - S(5) = \boxed{11.7503\%}.$$

Among 100 initially surviving issuers, expected one-year defaults are  $100(0.024690) = \boxed{2.469}$ . The spread approximation is

$$s \approx 0.025(0.60) = 0.0150 = \boxed{150 \text{ bps}}.$$

An observed spread may differ because it also reflects liquidity, risk premia, taxes, and uncertainty about recovery and intensity.

## 5 Lecture 11: Interest-Rate Options

### 5.1 Problem 17: Treasury Futures Put Quotation

A trader buys one 110 Treasury futures put for 0-48. Option premiums are quoted in 64ths, and one point is worth \$1,000.

- Convert the premium to points and dollars.
- Compute the expiration break-even price in decimal and fractional notation.
- Compute payoff and net profit if futures finish at 107-32.
- Compute payoff and net profit if futures finish at 112-00.

#### **i** Solution

The quote means 48/64 of one point:

$$0-48 = \frac{48}{64} = 0.75 \text{ point},$$

so the premium is  $0.75(\$1,000) = \boxed{\$750}$ . Break-even for a long put is

$$K - p = 110 - 0.75 = 109.25 = \boxed{109-16},$$

because  $0.25 = 16/64$ . At 107-32,  $F_T = 107.50$ , so

$$\text{payoff} = 110 - 107.50 = 2.50 \text{ points} = \$2,500,$$

$$\text{net profit} = 2,500 - 750 = \boxed{\$1,750}.$$

At 112-00 the put is out of the money, payoff is zero, and net profit is  $\boxed{-\$750}$ . It should not be exercised.

## 5.2 Problem 18: Put-Call Parity for Futures Options

European futures options have  $F_0 = 112$ ,  $K = 110$ ,  $r = 4\%$ ,  $T = 0.5$ , and call premium  $c = 3.10$  points.

- Compute  $e^{-rT}$ .
- Use  $c - p = e^{-rT}(F_0 - K)$  to compute the fair put.
- If the market put is 1.40, compute the mispricing in points and dollars.
- Describe an arbitrage that locks in the difference.

### Solution

$$e^{-rT} = e^{-0.04(0.5)} = \boxed{0.980199}.$$

Therefore,

$$p = c - e^{-rT}(F_0 - K) = 3.10 - 0.980199(2) = \boxed{1.1396 \text{ points}}.$$

The market put at 1.40 is overpriced by

$$1.40 - 1.1396 = 0.2604 \text{ point} = \boxed{\$260.40}.$$

At market prices, a long call and short put cost  $3.10 - 1.40 = 1.70$  points and have expiration payoff  $F_T - K$ . Add a short futures contract, whose payoff is  $F_0 - F_T$ , and the combined expiration payoff is the certain amount  $F_0 - K = 2$ . Borrow its present value, 1.9604, to finance the 1.70 option cost. The time-0 surplus is  $1.9604 - 1.70 = 0.2604$  point, while the expiration cash flows repay the borrowing.

## 5.3 Problem 19: Black-Scholes on a Zero-Coupon Bond

A European call and put are written on a zero-coupon bond with  $S_0 = 95$ ,  $K = 94$ ,  $T = 0.75$ ,  $r = 3\%$ , and bond-price volatility  $\sigma = 12\%$ .

- Compute  $\sigma\sqrt{T}$ ,  $d_1$ , and  $d_2$ .
- Use  $N(d_1) = 0.6444$  and  $N(d_2) = 0.6050$  to compute the call.
- Compute the put using complementary normal probabilities.
- Verify put-call parity.

#### Solution

$$\sigma\sqrt{T} = 0.12\sqrt{0.75} = \boxed{0.103923}.$$

$$d_1 = \frac{\ln(95/94) + (0.03 + 0.5(0.12)^2)(0.75)}{0.103923} = \boxed{0.3703},$$

$$d_2 = d_1 - 0.103923 = \boxed{0.2664}.$$

The call value is

$$c = 95(0.6444) - 94e^{-0.03(0.75)}(0.6050) = \boxed{5.61}.$$

Using  $N(-d_1) = 0.3556$  and  $N(-d_2) = 0.3950$ ,

$$p = 94e^{-0.0225}(0.3950) - 95(0.3556) = \boxed{2.52}.$$

The values satisfy parity up to rounding:

$$c - p \approx 5.61 - 2.52 = 3.09,$$

$$S_0 - Ke^{-rT} = 95 - 94e^{-0.0225} = 3.09.$$

## 5.4 Problem 20: Delta-DV01 Protective Put Hedge

A \$12 million bond portfolio has modified duration 7.50. A Treasury futures contract has DV01 \$75, current price 112, and point value \$1,000. A put has strike 111, premium 1.10 points, and delta  $-0.45$ .

- Compute portfolio DV01 and futures-equivalent contracts.
- Compute the number of puts for an initial delta-DV01 hedge and premium cost.
- If rates rise 50 bps and futures fall 3.75 points at expiration, compute the portfolio loss, put payoff, and net hedged P&L using a whole-number hedge.
- If rates instead fall 50 bps and the put expires worthless, compute net P&L.

## **i** Solution

Portfolio DV01 is

$$DV01_P = \$12,000,000(7.50)(0.0001) = \boxed{\$9,000}.$$

The portfolio is equivalent to

$$N_{fut-eq} = \frac{9,000}{75} = \boxed{120 \text{ futures contracts}}.$$

Because each put supplies only 0.45 short futures equivalent initially,

$$N_p = \frac{9,000}{0.45(75)} = 266.67,$$

so use 267 puts. Premium cost is

$$267(1.10)(\$1,000) = \boxed{\$293,700}.$$

For a 50 bp rate rise, the duration loss is

$$-\$9,000(50) = -\$450,000.$$

The expiration futures price is  $112 - 3.75 = 108.25$ . Each put pays  $111 - 108.25 = 2.75$  points, so total payoff is

$$267(2.75)(\$1,000) = \$734,250.$$

Net hedged P&L is

$$-450,000 + 734,250 - 293,700 = \boxed{-\$9,450}.$$

If rates fall 50 bps, the mapped portfolio gain is \$450,000. With no put payoff,

$$\text{net P\&L} = 450,000 - 293,700 = \boxed{\$156,300}.$$

The hedge is delta-neutral only for a small move when initiated. By expiration the put's nonlinear payoff can create gains or losses different from a futures hedge, and the premium is paid in either scenario.